



Are precious metals a hedge against exchange-rate movements? An empirical exploration using bayesian additive regression trees

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ABSTRACT

We use Bayesian additive regression trees to reexamine whether precious metals are a hedge against exchange-rate movements. We quantify the hedging properties of several major exchange rates, and we study how the marginal effects of appreciations/depreciations and across times of small/large exchange-rate movements. Results show that investments in gold and silver are strong hedges against major exchange rates. The hedging properties of palladium and platinum are confined to the Australian dollar and Canadian dollar. We also study whether investments are a safe-haven in times of large exchange-rate movements.

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1. Introduction

In recent years, significant research has been done to study whether investments in precious metals are a hedge against exchange-rate movements. An investor can use, for example, a gold investment as a hedge against exchange-rate movements if the dollar-denominated gold price increases in times of a dollar depreciation, that is, if the correlation between returns and exchange-rate movements is positive. [Beckers and Soenen \(1984\)](#) argue that a positive correlation between returns and exchange-rate movements can be hypothesized on theoretical grounds because low U.S. interest rates make a dollar investment less attractive while, at the same time, a zero-yield gold investment becomes more attractive. Recent empirical evidence lending support to the hedging hypothesis has been documented by [Lucey \(2013\)](#), [Joy \(2011\)](#) and [Reboredo \(2013a\)](#).¹

We use a machine learning algorithm known as Bayesian additive regression trees (BART, see [McCulloch, 1998](#)) to reexamine the hedging hypothesis. The BART algorithm renders it possible to build models that help to explore how precious metals returns are linked to movements in several major exchange rates. Building with BART uses Bayesian techniques to compute and additively combine regression trees. In each recursive binary splits to subdivide the range of exchange-rate movements into non-overlapping regions, the response of precious metals returns is then set to some region-specific constant (on

Breiman, Friedman, Oshen, & Stone, 1983). Precious metals returns of granular size can be modeled by recursive splits, and by combining several regression trees. Regression trees lend themselves to study the hedging properties of precious metals investments because regression trees capture in a natural way even complex nonlinearities in returns and exchange-rate movements. In addition, special techniques like partial dependence plots and variable importance have been developed for the analysis of regression trees that make it straightforward to study the hedging properties of precious metals and to quantify the relative importance of predictor variables. For a review of regression trees, see Hastie, Tibshirani, & Friedman (2009), page 351). The latter property is important for precious metals, since several exchange rates, and because we control for the impact of variables like oil-price movements, have been commonly studied in earlier research (Baur & McDermott, 2010, Beckmann & Czudaj, 2013a & Ciner et al., 2013). While regression trees have several interesting advantages, applications of regression trees in economics are scarce (for a useful introduction with a focus on economic applications, see Varian (2014)). In recent years, Malliaris (2015) use regression trees to study gold returns. However, they derive their empirical results from single regression trees. A drawback of single regression trees is that their hierarchical structure makes them highly sensitive to small changes in the data. The BART algorithm overcomes this drawback by using Bayesian techniques to combine several trees.

We study the hedging properties of four precious metals: gold, silver, platinum, and palladium. Since the hedging properties of precious metals is interesting because earlier researchers have shown that they cannot be considered as safe-haven assets (Batten, Ciner, and Lucey (2010) report that different macroeconomic factors are relevant for the movements of precious metals. Based on an analysis of spillover effects, Batten, Ciner, and Lucey (2015) argue that the movements of precious metals are only weakly integrated. Moreover, Lucey and Sile (2015) find that the three precious metals, gold, silver, platinum, and palladium are safe-haven assets during some periods of time when gold is not a safe-haven asset. Agyei-Ampomah, Gounopoulos, and Mazouz (2014) suggest that, while gold acts as a hedge for losses in the U.S. dollar issued by countries with serious debt problems, other metals may provide better protection against losses in the U.S. dollar markets in times of market jitters.

Adopting the standard definition of hedges and safe-haven investments proposed in earlier literature (Baur & McDermott, 2010), we find that gold and silver investments are a strong hedge against exchange rate movements of the U.S. dollar vis-à-vis the yen, the euro, the pound, the Canadian dollar, and the Australian dollar. Specifically, gold and silver investments exhibit a positive correlation with exchange-rate movements in times of U.S. dollar depreciations. While the positive correlation continues to hold and in some cases strengthens in times of large dollar depreciations, gold and silver investments act as safe-haven investments in times of large depreciations. It also holds for platinum and palladium, as documented by Wang and Lee (2011), who report that gold investments are an effective exchange rate hedge against large depreciations of the yen. The exchange-rate-hedging property of palladium and platinum is also documented for the Canadian dollar and the Australian dollar. In quantitative terms, both the Canadian dollar and the Australian dollar, two classic “commodity currencies”, tend to have a higher explanatory power for precious metal returns than the U.S. dollar exchange rates (on commodity exchange rates, see Chen, Rogoff, & Rossi (2010)).

We further find that an investment in gold acts as a strong hedge against U.S. stock-market downturns (Baur & McDermott, 2010). Silver returns, in contrast, show no systematic response to stock-market movements. Platinum and palladium investments are only a hedge against stock-market movements. Further, gold returns exhibit a positive correlation with oil-price increases, which is in line with results reported by Reboredo (2013b). Platinum and palladium returns, in turn, tend to react procyclically to stock-market movements, possibly reflecting fluctuations in the U.S. business-cycle frequencies. Price dynamics, thus, substantially differ across precious metals, lending support to the view that they cannot be considered as a single asset class (Batten et al., 2010, 2015). All four precious metals return series are modeled by a nonlinear way to commodity-market movements as represented by returns on the widely-traded Goldman Sachs Commodity Index (GSCI).

We proceed as follows. In Section 2, we lay out the BART algorithm. In Section 3, we describe our empirical results. In Section 4, we summarize our empirical results. In Section 5, we conclude.

2. Bayesian additive regression trees

A regression tree uses binary recursive splits to partition the space of the predictors into non-overlapping regions. The binary splits imply that a regression tree consists of a root, interior nodes, and terminal nodes. The distance from the root to the terminal nodes (the “leaves”). We further define $M = (\mu_1, \dots, \mu_M)$ with M denoting the number of terminal nodes. Every leaf parameter is the best prediction of precious metal returns in the region defined by the leaf. The BART algorithm combines several regression trees to

modeling is to build a flexible prediction model for precious metals returns, y , using an ensemble of m following format:

$$y_t = \sum_{j=1}^m f(x|T_j, M_j) + \epsilon_t, \quad \epsilon \sim N(0, \sigma^2),$$

where ϵ is a normally distributed disturbance term. The function f relates the leaf parameters of the predictors, x .

Eq. (1) highlights the difference between BART modeling and a widely-studied modeling approach of the hedging and safe-haven property of precious metals. Baur and Lucey (2010), for example, consider the following regression model:

$$y_t = b_0 + b_1 x_t + b_2 x_{t,q} + \epsilon_t,$$

where $b_j, j = 0, 1, 2$ denotes parameters to be estimated, and $x_{t,q}$ denotes the q -th quantile of some predictor. b_1 captures the average dependence of y_t on x_t , while the parameter b_2 captures the nonlinear dependence of extreme market movements. When the predictors being studied are stock or bond returns, then either b_1 implies that y_t and x_t are uncorrelated or negatively correlated on average such that y_t is a hedge. The sum of $b_1 + b_2$ is non-positive then y_t is a safe haven because y_t and x_t are uncorrelated or negatively correlated when x_t assumes extreme values.

Eq. (2) requires that a researcher specifies the specific values of the quantiles, q , a choice which is arbitrary. The choice of the quantiles becomes even more involved in case of exchange-rate movements, wanting to study separately gold returns in times of both large appreciations and large depreciations. Moreover, to keep the model parsimonious then Eq. (2) leaves potentially important complex nonlinearities undetected. The regression model given in Eq. (2) becomes rather convoluted once a researcher studies, as we do in this paper, the relation of y_t with several predictors across various quantiles. Yet another feature of Eq. (2) is that it invokes a linear structure that rules out, in a multivariate context, interaction effects between different predictors. Interaction effects arise when the dependence of, for example, gold returns on stock returns depends on the bond market. BART modeling provides a unified platform that renders it possible to model even nonlinearities and interaction effects between predictors and, in addition, allows the relative importance of predictors to be studied.

BART modeling requires incorporating prior information on the tree structure (T_j, M_j) and the regression coefficients to achieve a sufficient amount of regularization.² Like Chipman et al. (1998), we set the prior for T_j , which consists of the interior nodes in a tree, equal to $\alpha(1+d)^{-\beta}$, where $\alpha \in (0, 1)$ and $\beta \in [0, \infty)$. We fix α and β to assume 0.95 and 2, implying that moderately-sized trees with 2 or 3 nodes have the highest likelihood, and a number of leaf parameters.³ The prior on each leaf parameter is assumed to be normally and identically distributed, $\mu \sim N(\mu_\mu/m, \sigma_\mu^2)$, and the expected value of μ_μ is $(y_{\max} + y_{\min})/2$. We further assume an Inverse Gamma distribution for the residual variance, $\sigma^2 \sim IG(v/2, v\lambda/2)$. The parameter λ is estimated empirically from the data such that the q -percentage chance to improve the root mean squared error (RMSE) of a least squares regression. Furthermore, we prevent σ^2 from getting too small and reduces the probability of over-fitting. Finally, the parameters, μ_μ and σ_μ , are assumed to be normally distributed, $\mu_\mu \sim N(m\mu_\mu - k\sqrt{m}\sigma_\mu, m\sigma_\mu^2)$ and $\sigma_\mu \sim N(m\sigma_\mu + k\sqrt{m}\sigma_\mu, m\sigma_\mu^2)$. The parameter k controls for the tightness of the prior. A large value of k leads to a stronger regularization of μ_μ .

Equipped with the priors, BART modeling uses an iterative Bayesian MCMC backfitting algorithm to approximate the posterior distribution of the sum-of-trees function in Eq. (1). Chipman et al. (2010) discuss in depth the iterative Bayesian MCMC backfitting algorithm, and Kapelner and Bleich (2016) discuss its implementation. The algorithm follows: The first step is to fit every individual tree, T_j , given all other trees in the ensemble, to the partial residuals when the j -th tree is excluded, where a Metropolis-Hastings algorithm is used to propose (and accept) a new tree structure. Once equipped with the new structure, the algorithm samples in the second step from the posterior distribution of the parameters, completing the new estimate of T_j . The first two steps are then repeated in an iterative manner for all trees in the ensemble using the respective updates of the partial residuals. In a final step, once the ensemble is updated, Eq. (1) is used to extract the residuals of the BART model, which then can be used to sample from the posterior distribution of the variance parameter, σ^2 . Going through these steps over and over again gives a large number of

3. The data

Like [Reboredo \(2013a\)](#), we study weekly precious metals returns (last trading day of a week). We use the first-differenced natural logs of the prices of the four precious metals being studied. The sample period runs from January 1990 to December 2015, where the data sources are the FRED board of the Fed of St. Louis and Datastream. All prices are in US dollars (metal prices (in natural logs)).

The gold price showed a slight upward trend at the beginning of the sample period. This upward trend accelerated around 2005 and reached its peak in spring 2011. Until around April 2012 the gold price experienced both up- and downward movements, and then it started declining. The development of the silver price is closely related to the development of the gold price, where an unsettled question is whether the gold and silver prices move in tandem ([Baur & Tran, 2014](#); [Pierdzioch, Risse, & Rohloff, 2015](#)). The silver price exhibited an upward movement throughout the sample period and reached a strong peak in April 2011. Thereafter, it started declining again.

As compared to the prices of gold and silver, the prices of the other two precious metals, platinum and palladium, followed a rather different development throughout the sample period. The price of platinum showed a steady upward trend from the beginning of the sample period and peaked in April 2008. Thereafter, the price of platinum, but also the price of palladium, experienced a sharp drop that resulted in a price decline of more than 60% between June and October 2008. This decline is reflected in the annual platinum and palladium reports published by Johnson Matthey (Platinum 2008, 2009)⁵ the cause of which was mainly caused by heavily decreasing outputs in the two main platinum-producing countries, South Africa and Russia, which were caused by local problems like shutdowns due to violated safety instructions and strikes. At the same time, platinum and palladium demand rose. According to the 2009 report, the sharp decline between 2007 and 2008 was a result of the global economic crisis. Many platinum- and palladium-demanding firms reduced their production, which led to a severe decline in demand and tumbling prices.

[Table 1](#) shows summary statistics of precious metals returns. As expected, mean returns are positive for all metals. The standard deviation of returns is smallest for gold returns and largest for palladium returns. A negative skewness and excess kurtosis are characteristic features of returns, albeit to different extents across the precious metals. None of the tests can be rejected at conventional significance levels. Gold returns show a somewhat stronger first-order autocorrelation than returns of the other precious metals. For estimation of the BART models, we have a total of 885 observations for every precious metal in our sample.

In order to analyze the hedging performance of the four precious metals, our list of predictors comprises the weekly returns of the exchange rate of the dollar vis-à-vis the yen, the euro, the British pound, the Canadian dollar, the Australian dollar, and an aggregate trade-weighted exchange rate index (the Trade-Weighted Exchange Index of the Fed, TWEXB, available as Wednesday data). In addition, we control for the important macroeconomic variables that have been widely studied in earlier research:⁶ weekly returns of the S&P-500 index, the return of the commodity-price index, the Chicago Board Options Exchange Volatility Index (VIX), the term spread (the difference between the 3-month treasury constant maturity rate minus the 3-months treasury constant maturity rate), and the corporate bond spread (computed as Moody's Baa rate minus Moody's Aaa rate).

4. Empirical results

4.1. Model calibration and performance

We compute all empirical results documented in this research using the free R programming environment ([R Core Team, 2015](#)). For estimation of the BART models, we use the add-on package “bartMachine” ([Bleich, 2016](#)). We use 7,000 runs of the Bayesian MCMC backfitting algorithm and discard the first 2,000 iterations. For the calibration of the parameters for prior selection we mainly use the default parameters recommended by [Chipman et al. \(2010\)](#): $m = 50$, $\alpha = 0.95$, $\beta = 2$, $v = 3$, $q = 0.9$, $\kappa = 5$.⁷

While the default calibration is a reasonable benchmark, we also inspected whether the results are robust to variations in the parameter values. To this end, we used a grid-search approach. The grid-search approach searches the range of parameter values recommended by [Chipman et al. \(2010, Table 2\)](#): $m = \{50, 100, 200\}$, $(v, q) = \{(3, 0.9), (3, 0.99), (10, 0.75)\}$. Using five-fold cross validation, the optimal parameter combination is the one that produces the lowest out-of-sample RMSE. Results of the grid-search approach (not reported) show that the results reported in this research are robust to variations in the parameter values.

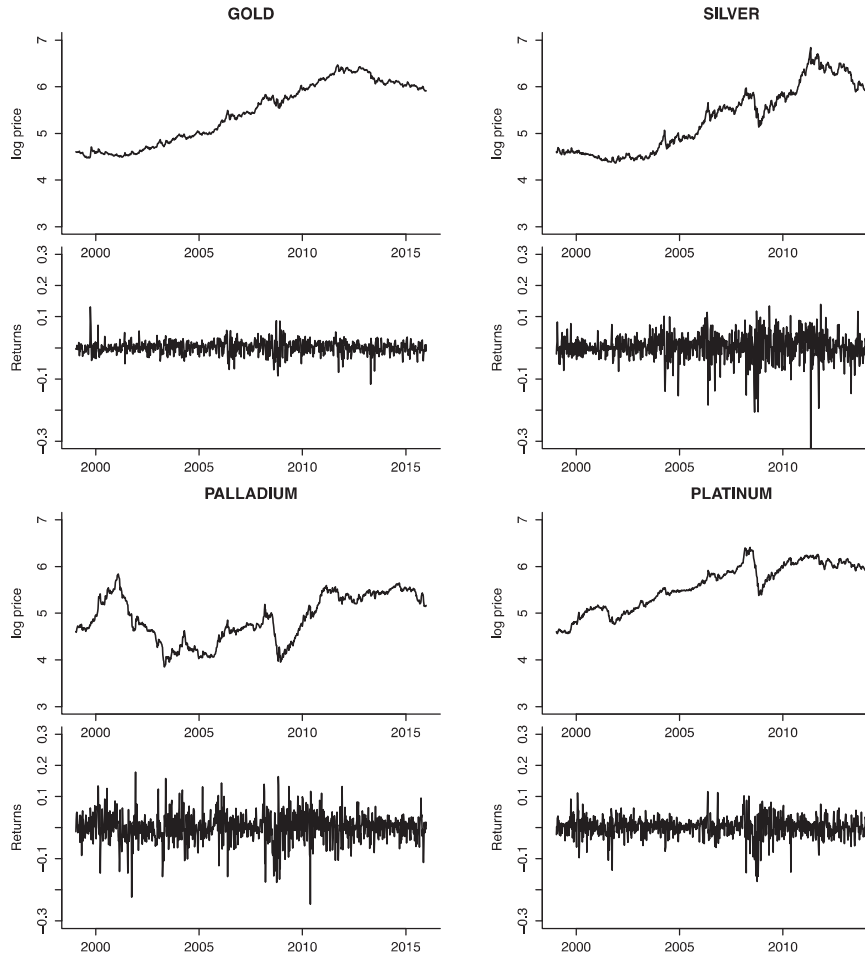


Fig. 1. Precious metals prices. *Note:* Precious metals prices are in natural logs. Anti-logs were scaled to assume the value 100 at period.

Table 1
Descriptive statistics of returns.

Statistic	Gold	Silver	Palladium
Mean	0.0015	0.0011	0.0006
SD	0.0211	0.0433	0.0472
Max	0.1313	0.1398	0.1790
Min	−0.1166	−0.3535	−0.2459
0.25 Quantile	−0.0102	−0.0206	−0.0245
0.75 Quantile	0.0132	0.0257	0.0248
Skewness	−0.1407	−1.1830	−0.3871
Kurtosis	7.0429	10.3606	5.7919
AR(1)	0.2095	−0.0196	0.0738
Observations	885	885	885

Note: SD = standard deviation. AR(1) = autocorrelation of order one.

Table 2
Full-sample performance.

Statistic	Gold	Silver	Palladium
<i>RMSE</i>	0.0176	0.0369	0.0413
95% CI (lower)	0.0171	0.0360	0.0405
95% CI (upper)	0.0181	0.0377	0.0422
R^2	0.3078	0.2800	0.2330
95% CI (lower)	0.2728	0.2460	0.2000
95% CI (upper)	0.3469	0.3155	0.2625
<i>AUROC</i>	0.7799	0.7155	0.6836
95% CI (lower)	0.7616	0.6958	0.6620
95% CI (upper)	0.7974	0.7342	0.7037

Notes: *RMSE*: Root mean-squared error computed as the mean across the simulated posterior samples. R^2 : Pseudo coefficient of determination computed as the mean across the simulated posterior samples. *AUROC*: Area under the receiver-operating characteristics curve computed as the mean across the simulated posterior samples. 95% CI (lower, upper): Lower and upper boundary of the statistics.

Table 3
Out-of-sample performance.

Statistic	Gold	Silver	Palladium
BRT	0.2287	0.1605	0.1230
BART	0.2220	0.1775	0.1584

Notes: R^2 : Out-of-sample pseudo R^2 statistic computed following the approach proposed by [Campbell and Thompson \(2008\)](#). The pseudo R^2 statistic compares the out-of-sample mean to the historical in-sample mean. The in-sample period includes data up to and including December 2006. The BRT models were estimated using the *gbm* package in R with the following parameters: a learning rate of 0.005, a bag fraction of 0.5, a minimum number of observations per terminal node of 5, and a tree depth of 5. The number of trees was determined using five-fold cross-validation. The BRT models were estimated using the R package “gbm”.

Table 3 summarizes the out-of-sample performance of the BART models. In order to study out-of-sample performance, we estimate the BART models using data on a subsample period comprising data up to and including December 2006. We then compute out-of-sample forecasts and use the out-of-sample forecasts to compute a pseudo R^2 statistic. The pseudo R^2 statistic is suggested by [Campbell and Thompson \(2008\)](#). The pseudo R^2 statistic compares the out-of-sample BART model to a benchmark mark model that uses the historical in-sample mean to forecast precious metals returns. A positive pseudo R^2 indicates that the BART model outperforms the benchmark model. We also compute a pseudo R^2 statistic using the BRT model. The BRT model (trees (BRT, see [Friedman, 2001](#)), a close competitor of BART modeling. A BRT model also computes forecasts of precious metals returns from an ensemble of trees, but rather than relying on Bayesian backfitting it uses boosting. Boosting combines regression trees (for details, see also [Chapter 10](#) [Hastie et al., 2009](#)).

The punchline of **Table 3** is that, although details differ across the precious metals under consideration, the out-of-sample performance is not too different across the BART and the BRT models.⁸ An advantage of BART over BRT is that BART modeling provides a natural platform for significance testing. Significance testing is useful, for example, to test the effects of the predictors on precious metals returns (see [Section 4.3](#)), but also for testing whether a specific predictor has significant predictive power for precious metals returns. For the purpose of our study, it is useful to test the null hypothesis that exchange-rate movements jointly have no effect on precious metals returns. To this end, we use a permutation test ([Kapelnier & Bleich, 2016](#)). The permutation test “destroys”, in a first step, any link between exchange-rate movements and exchange-rate movements by permuting the columns containing the exchange-rate movements in the predictor matrix. In a second step, a BART model is estimated using the permuted predictor matrix and a pseudo R^2 statistic is computed. The sampling distribution of the pseudo R^2 statistic under the null hypothesis is simulated by repeating the steps several times. The null hypothesis can then be tested by comparing the sampling distribution of the pseudo R^2 statistic computed under the null hypothesis with the pseudo R^2 statistic of the original BART model. **Table 4** shows the results of the permutation test.

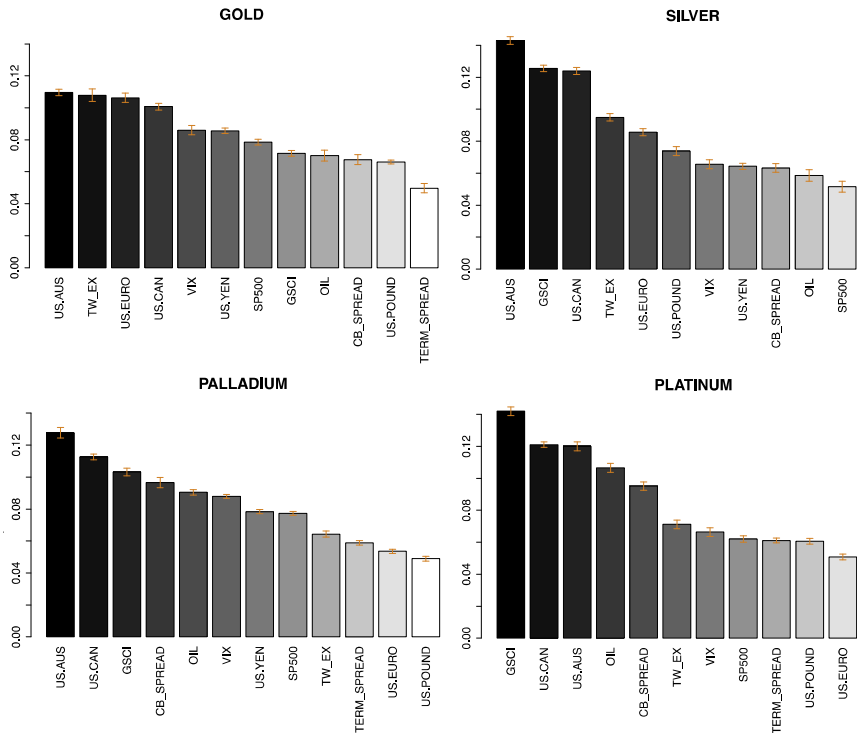
The null hypothesis is strongly rejected for all four precious metals. Hence, exchange-rate movements have significant predictive power for precious metals returns. The natural next steps, thus, are to assess the relative importance of exchange rates ([Section 4.2](#)) and to dissect how exactly precious metals returns fluctuate when exchange rates change ([Section 4.3](#)).

Table 4

Joint test of predictive power of exchange-rate movements.

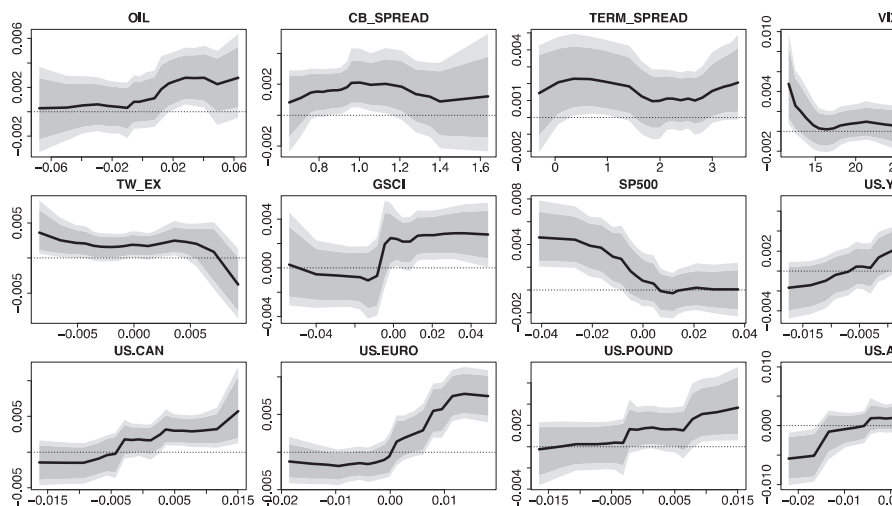
Statistic	Gold	Silver	Palladium
<i>p</i> -Value	<0.0001	<0.0001	<0.0001

Notes: The table summarizes the *p*-values of a permutation test of the null hypothesis that the exchange-rate movements have no predictive power for precious-metal returns. Number of simulation runs: 1000. The *p*-values are defined as the proportion of pseudo R^2 statistics that exceed the pseudo R^2 statistic implied by the original BART model.

**Fig. 2.** Variable importance.

Results show that the exchange rates vis-à-vis the Australian and Canadian dollar are among the top predictors for gold and silver returns. The importance of other exchange rates, in contrast, varies across the precious metals. For gold and silver, the exchange rate vis-à-vis the euro is more important than the exchange rate vis-à-vis the Canadian dollar. The euro exchange rate is somewhat less important in the case of silver returns than in the case of gold returns, and it is not used often as a splitting variable in the cases of palladium and platinum. As far as palladium and platinum returns are concerned, the corporate-bond spread and oil-price movements are more important than the exchange rates. The term spread is somewhat more important for explaining palladium and platinum returns than for gold and silver returns, but it is not among the top predictors. S&P 500 returns are less important for explaining gold and silver returns, while the estimated BART model finds a somewhat larger intermediate importance of stock returns for gold and palladium returns. Interestingly, GSCI returns are more important for explaining silver, palladium, and platinum returns than for explaining gold returns. Gold, thus, seems to be to a larger extent a financial asset than the other three precious metals, decoupled from price developments in commodity markets than the other three precious metals in general. Before jumping to conclusions, it is interesting to use partial dependence plots to explore in detail how the returns of the precious metals under consideration depend on the various predictors.

Panel A: Gold



Panel B: Silver

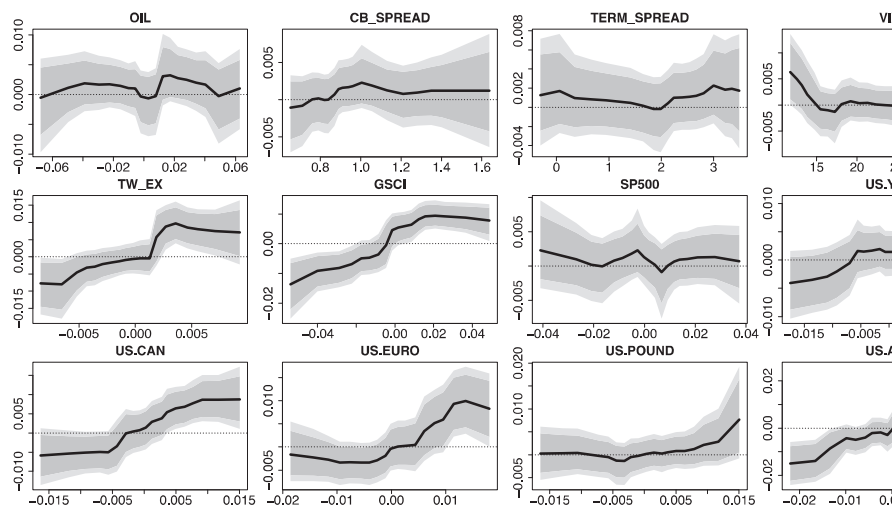
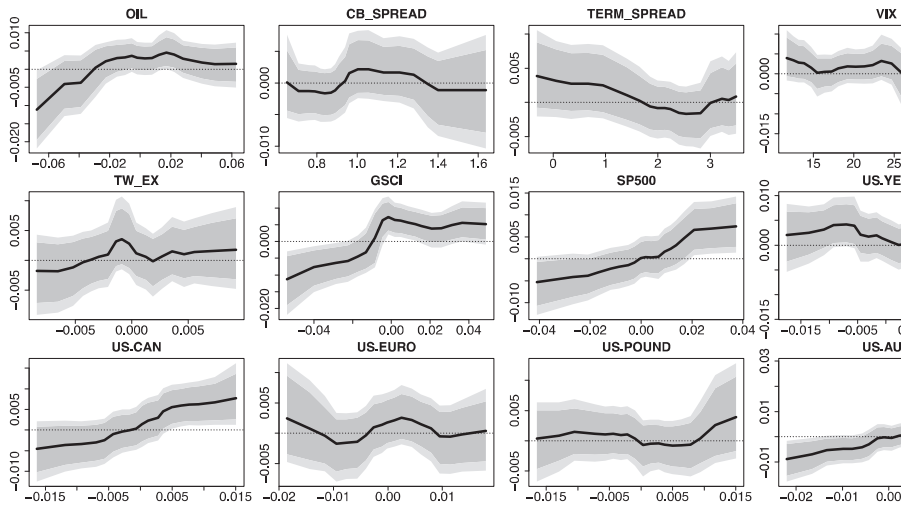


Fig. 3. Partial dependence plots (Part I).

turn, has a positive slope and is significantly different from zero. A precious metal is a safe-haven in times of market stress characterized by large depreciations, even if it is uncorrelated with exchange-rate movements in certain periods of time. The partial dependence plot should be significantly different from zero in times of large exchange rate movements.

The marginal effects reveal that gold and silver returns turn significantly positive in case of dollar depreciations. That both gold and silver investments are strong hedges against dollar depreciations. For example, a depreciation of the dollar vis-à-vis the Canadian dollar by 0.5% is accompanied, on average and holding all other predictors constant, by a 0.5% increase in gold and silver returns.

Panel A: Palladium



Panel B: Platinum

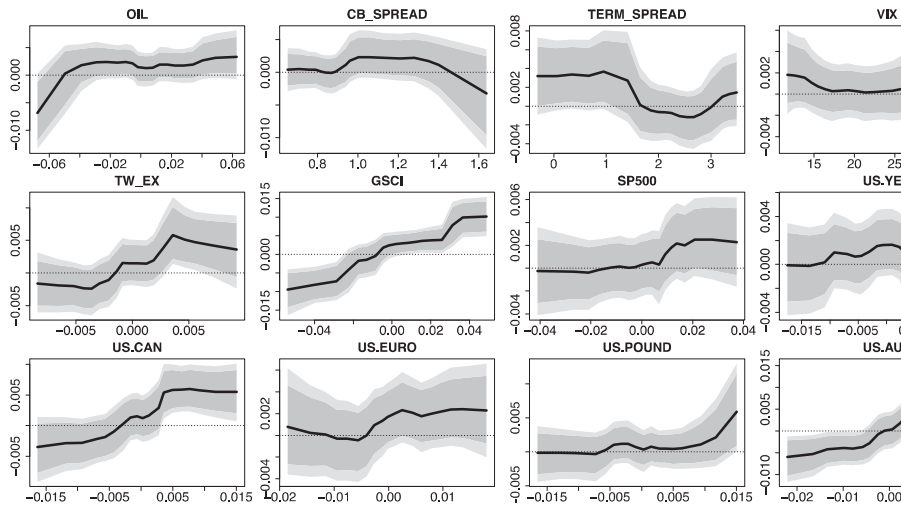


Fig. 4. Partial dependence plots (Part II).

(safe-haven) property holds in case of palladium and platinum with respect to large depreciations of the Canadian dollar and the Australian dollar. There is weak evidence that the safe-haven property of platinum investments also holds with respect to appreciations vis-à-vis the Australian dollar.

As for the aggregate trade-weighted exchange rate, the marginal effects for gold indicate a counter-cyclical response. The other three precious metals, in contrast, exhibit a procyclical response, which is also evident in the cases of silver returns and, to a lesser extent, platinum returns.

Correlation results reported by [Wang and Lee \(2011\)](#) it is also evident from the marginal effects

gold returns become positive when the S&P-500 returns are negative, confirming that gold investm against U.S. stock-market declines (see also [Baur & McDermott, 2010](#)). The marginal effects also visual linear transition between the response of gold returns to small U.S. stock-market declines (“hedging pe to large stock market declines (“safe-haven” times; see also [Beckmann, Czudaj, & Pilbeam, 2015b](#)). In ment is a weak hedge against fluctuations in the S&P-500 because silver returns show no systematic r movements. Platinum and especially palladium returns, in turn, show a procyclical reaction to stock-ma procyclical reaction results in a less than perfect correlation, palladium and platinum are diversifier market movements. A precious metal is a portfolio diversifier if it exhibits on average a less than per with another asset ([Baur & Lucey, 2010](#)).

Finally, gold returns exhibit a positive marginal effect with respect to oil-price increases, corrobor mented by [Reboredo \(2013b\)](#). This effect, however, is only borderline significant. Palladium and plat bit a procyclical marginal effect with respect to oil-price movements, but again this effect is hardly

4.4. The recent financial crises

A natural question is whether the marginal effects with respect to exchange-rate movements sum have been stable over time. In particular, it is interesting to study the stability of the marginal effect financial market crises and financial market turmoil. Available empirical evidence is mixed in this r reports that gold investments have become an increasingly effective hedge against dollar currency r [Zagaglia and Marzo \(2013\)](#) using a different empirical technique suggest that the comovement of g fluctuations has been stable during recent periods of financial market jitters. [Capie et al. \(2005\)](#) doe investments were a hedge against fluctuations of key dollar exchange rates, its hedging property u time due to shifts in political attitudes and unpredictable events.

[Figs. 5 and 6](#) plot recursively estimated marginal effects. In order to produce the recursive estimat BART models using data up to and including 2007 using the model calibrations given in Section 4.1. V

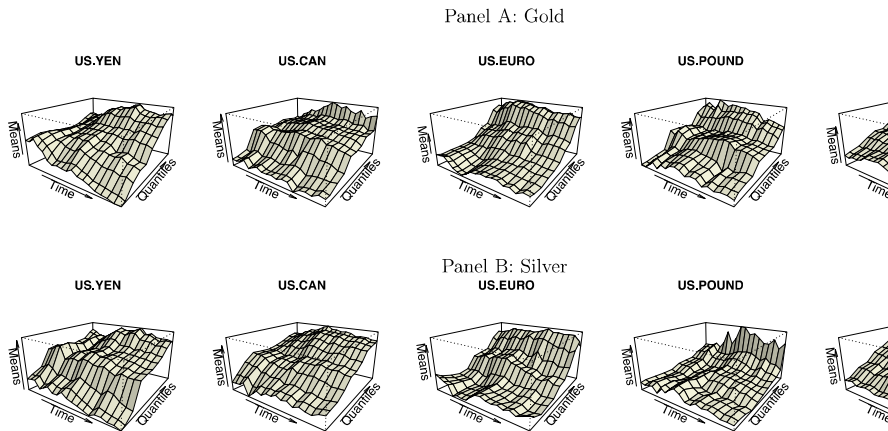
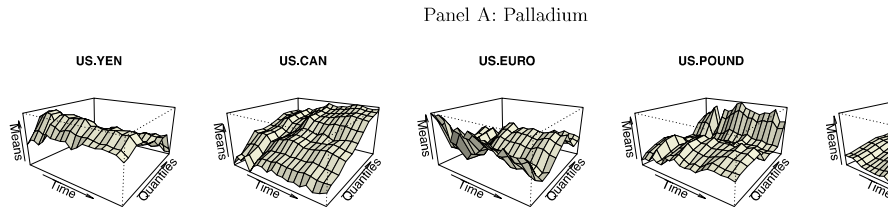


Fig. 5. Recursive posterior mean partial dependence (Part I).



six months of data and reestimate the models. In general, results show that the marginal effects have over time. In the cases of gold and silver, appreciations of the dollar vis-à-vis the yen and the Australian dollar accompanied in recent years by stronger negative returns. As a result, the response of gold and silver in the yen and the Australian dollar have become more symmetric during the recent financial crises.

5. Concluding remarks

Bayesian additive regression trees (BART) are a powerful technique to study the hedging properties of precious metal investments. Tools for the visualization of predictor importance and marginal effects make it straightforward to interpret BART estimation results in economic terms.

Our estimation results show that gold and silver investments are a strong hedge against depreciations of the dollar vis several major exchange rates. Palladium and platinum investments are a strong hedge mainly with respect to the Canadian dollar and Australian dollar, while such investments in general are weak hedges against the yen and other exchange rates. We also have used our estimation results to shed light on the safe-haven properties of precious metal investments. For all four precious metals the importance as a splitting variable of returns of both the yen and the Australian dollar, two “commodity currencies”, dominate the importance of the other exchange-rate returns.

We also have documented asymmetries and nonlinearities in the marginal effects with respect to precious metal investments. Our results show that researchers and portfolio managers should take into account the possible asymmetries and nonlinearities are characteristic features of the comovement of precious metal returns and exchange rates movements. On a more general level, our results indicate that asymmetries and nonlinearities matter for the modeling of the interplay and potential causal links between other commodity prices and exchange rates. As far as “commodity currencies” are concerned. While nonlinearities have been analyzed in earlier research on the link between oil prices and exchange rates (Beckmann & Czudaj, 2013b), our results show that BART is a flexible platform that can recover even complex nonlinearities that may be difficult to detect by means of traditional modeling techniques. In a similar vein, our results show that BART modeling lends itself to the study of potentially asymmetric links between commodity prices and the exchange rates of “commodity currencies” during periods of appreciations and depreciations.

Our results further show that all four precious metals react in a procyclical but nonlinear way to changes in exchange rates, and gold investments further act as a strong hedge against U.S. stock-market downturns. In particular, and especially palladium, which are mainly used as special production materials in certain industries, store of value by investors, react procyclically to stock-market fluctuations.

BART modeling lends itself to study at least three important research questions in future research. First, we can adapt BART modeling to explore the links between other commodity prices and exchange-rate and financial market movements. In this respect, it is interesting not only to analyze the comovement of commodity prices and exchange rates but also to use BART modeling to shed light on whether commodity prices help to forecast exchange rates (Beckmann et al., 2014). Second, while we have focused on dollar exchange rates, it is interesting to study the links between precious metals returns and, on the other hand, cross-rate movements and movements in international stock markets. It is interesting to quantify the extent to which techniques like partial dependence plots and variable importance measures can help investors to improve portfolio-allocation decisions.

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